



Investigation of Current-Phase Relationship in 1T-Phase Platinum Ditelluride Josephson Junctions under In-Plane Magnetic Field

面內磁場下 1T 相二碲化鉑約瑟夫森接面之電流-相位關係研究

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Introduction

Outline

Introduction	Theory	Methods	Results
• Research motivation • 1T-PtTe₂ (Type-II Dirac)	• Josephson effect & CPR • Josephson diode effect	• Device fabrication • Measurement setup	• Basic characterization • CPR analysis • In-plane field effect

Research Motivation

Topological SC	CPR	Diode Effect
• Type-II Dirac semimetal PtTe₂ proximitized by a superconductor	• Direct probe of junction physics beyond simple $\sin \phi$	• Non-reciprocal transport $I_c^+ \neq I_c^-$ • Route to dissipationless

Theoretical Background

Josephson Effect & Current – Phase Relation

Josephson relations

- DC: $I_s = I_c \sin \phi$ (ideal tunnel junction)
- AC: $\dot{\phi} = \frac{2e}{\hbar} V$

Josephson Diode Effect (Concept)

Definition

- Non - reciprocal critical current
 $I_c^+ \neq |I_c^-|$
- Superconducting rectification without Joule heating

Beyond sinusoidal CPR

- High transparency $\rightarrow$ skewed CPR
- Harmonics:
$$I(\phi) = \sum_n I_n \sin(n\phi + \delta_n)$$
- Access via **asymmetric SQUID** readout

Typical ingredients

- Broken inversion symmetry + broken time - reversal symmetry
- Often tuned by $B_{\parallel}$ with strong spin – orbit coupling

Device Fabrication Flow

1. **Crystal Growth:** Self-flux method (provided by Prof. C.S. Lue, NCKU).
2. **Exfoliation:** Mechanical exfoliation of PtTe_2 flakes onto SiO_2/Si substrates.
3. **E-beam Lithography (EBL):**
 - PMMA resist coating.
 - Pattern definition for contacts and SQUID loops.
4. **Metallization:**
 - Superconducting electrodes: Al (Aluminum) or Nb (Niobium).
 - In-situ Ar ion milling to ensure transparent interfaces.
 - E-beam evaporation & Liftoff.
5. **Design:** Asymmetric dc-SQUID geometry for CPR retrieval.

Measurement Setup

- **Cryogenic system:** Dilution refrigerator
(base ~ 40 mK)



Results: Basic Characterization

I-V Characteristics (DC Transport)

- Typical IV Curve:
 - Clear Superconducting branch ($V = 0$).
 - Switching Current (I_c) and Retrapping Current (I_r).
 - Hysteretic behavior (underdamped junction).

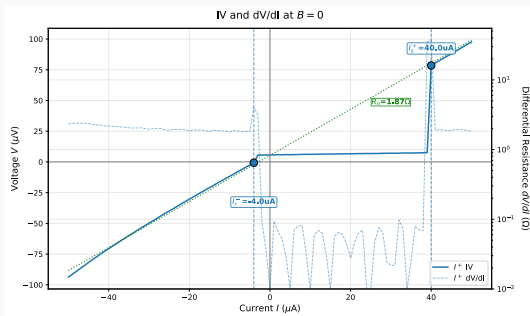


Figure 3: I-V Curve at Base Temp.

- Observation:
 - Sharp transition indicates high-quality junction.
 - Finite resistance in normal state (R_N).
 - Excess Current observed at high bias $\rightarrow$ High transparency interface.

Results: Current-Phase Relation

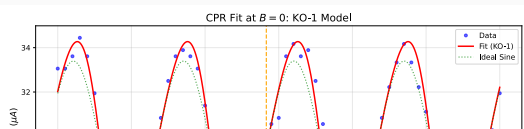
CPR Retrieval Method

- **Asymmetric SQUID Technique:**
 - Large junction (I_{ref}) // Small junction (I_{test}).
 - $I_c(\Phi_{ext}) \approx I_{ref} + I_{test}(\phi)$.
 - Assumes $I_{ref} \gg I_{test}$ and sinusoidal $I_{ref}(\phi)$.
- **Math Model:**

$$I_s(\phi) = I_1 \sin(\phi) + I_2 \sin(2\phi + \delta)$$

- I_1 : Fundamental component (conventional).
- I_2 : Second harmonic (4π -periodic / ABS).

Measured CPR & Fitting



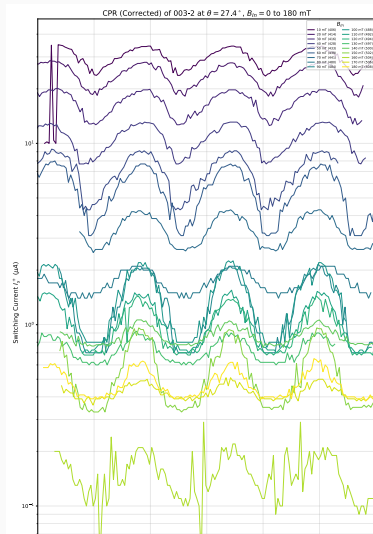
• Key Findings:

- **Skewed CPR:** Deviation from pure sine wave.
- **High I_2 / I_1 ratio:** Significant second

Results: In-Plane Magnetic Field Effect

Tunable CPR with In-Plane Field

- Applying $B_{\parallel}$ modifies CPR shape
- Suggests a **phase shift** (ϕ_0) in higher harmonics
- Correlates with emergence of the Josephson diode effect



Summary

1. Successful Fabrication:

- High-quality 1T-PtTe₂ Josephson junctions and SQUIDs realized.

2. Anomalous CPR:

- Observed significant **second harmonic** (I_2) contribution.
- Direct evidence of high-transparency transport modes.

3. Magnetic Tuning:

- In-plane magnetic field effectively modulates the CPR and induces **Josephson Diode Effect**.
- Demonstrates potential for **tunable superconducting logic** and **phase batteries**.

Future Outlook

- **Optimization:** Improving fabrication yield and interface control.
- **Material Exploration:** Extending to other Weyl/Dirac semimetals.
- **Quantum Information:** Integration into superconducting qubit circuits (e.g., as

